Unit: Unit 9: Data Statistical Analysis **Name:** ________________
Topic: Box Whisker (Dispersion) **Date:** __________

Advanced (Habanero)

Q1. What is the main purpose of a box-and-whisker plot?
(A) To display the mean of a dataset
(B) To compare the dispersion of two datasets
(C) To show the frequency of each data point
(D) To identify outliers in a dataset
(E) To calculate the standard deviation

Q2. Which part of a box-and-whisker plot represents the interquartile range?
(A) The whiskers
(B) The box
(C) The median line
(D) The outliers
(E) The axes

Q3. What is the Q_1 value in a dataset with the following ordered values: $2, 4, 6, 8, 10$?
(A) 4
(B) 6
(C) 8
(D) 2
(E) 10

Q4. A dataset has the following values: $1, 1, 2, 3, 5, 8, 13$. Which value is an outlier?
(A) 1
(B) 2
(C) 3
(D) 5
(E) 13

Q5. What is the IQR for a dataset with $Q_1 = 3$ and $Q_3 = 7$?
(A) 2
(B) 4
(C) 5
(D) 10
(E) 4

Q6. Which of the following is NOT a characteristic of a box-and-whisker plot?
(A) It displays the mean of the data
(B) It shows the median of the data
(C) It identifies outliers
(D) It displays the range of the data
(E) It is used for categorical data

Q7. A box-and-whisker plot has a median of 6 and an IQR of 4. If $Q_1 = 4$, what is Q_3?
(A) 6
(B) 8
(C) 10
(D) 12
(E) 8

Q8. What can be inferred about a dataset with a large IQR?
(A) The data is closely clustered around the mean
(B) The data is spread out over a large range
(C) The data has many outliers
(D) The data is symmetric
(E) The mean is greater than the median

Open Ended Questions (FRQ)

Q9. Construct a box-and-whisker plot for the following dataset: $2, 4, 6, 8, 10, 12, 14$. Identify the median, Q_1, Q_3, and IQR. Check for outliers.

Q10. Compare the dispersion of two datasets using box-and-whisker plots. Dataset 1: $1, 3, 5, 7, 9$. Dataset 2: $2, 4, 6, 8, 10, 20$. Which dataset has greater dispersion?

Chef's Tips

- Tip 1: Ensure students understand the concept of quartiles and interquartile range
- Tip 2: Emphasize the importance of identifying outliers in a dataset
- Tip 3: Encourage students to use box-and-whisker plots to compare the dispersion of different datasets